

Manuscript Title

Effects of weighted sled towing with heavy versus light load on sprint acceleration ability

Running Head

Effects of weighted sled towing

Laboratory Where the Research Was Conducted

School of Exercise and Health Sciences
Edith Cowan University
Joondalup, Western Australia, Australia

Authors

Naoki Kawamori^{1,2}

Robert U. Newton¹

Naruhiko Hori³

Kazunori Nosaka¹

Institutions

¹School of Exercise and Health Sciences
Edith Cowan University
Joondalup, Western Australia, Australia

²Japan Institute of Sports Sciences
Tokyo, Japan

³Athlete and Coach Services
Western Australian Institute of Sport
Claremont, Western Australia, Australia

Corresponding Author:

Naoki Kawamori, Ph.D., C.S.C.S.
Japan Institute of Sports Sciences
3-15-1, Nishigaoka
Kita-ku, Tokyo 1150056 Japan
Phone #: +81-3-5963-0251
Fax #: +81-3-5963-0252
E-mail: kawamori.naoki@jiss.naash.or.jp

ABSTRACT

Weighted sled towing is used by athletes to improve sprint acceleration ability. The typical coaching recommendation is to use relatively light loads, as excessively heavy loads are hypothesized to disrupt running mechanics and be detrimental to sprint performance. However, this coaching recommendation has not been empirically tested. This study compared the effects of weighted sled towing with two different external loads on sprint acceleration ability. Twenty-one physically active men were randomly allocated to heavy- ($n = 10$) or light-load weighted sled towing ($n = 11$) groups. All subjects participated in two training sessions per week for 8 weeks. The subjects in the heavy and light groups performed weighted sled towing using external loads that reduced sprint velocity by $\sim 30\%$ and $\sim 10\%$, respectively. Before and after the training, the subjects performed a 10-m sprint test, in which split time was measured at 5 and 10 m from the start. The heavy group significantly improved both the 5- and 10-m sprint time by $5.7 \pm 5.7\%$ and $5.0 \pm 3.5\%$, respectively ($P < 0.05$), whereas only 10-m sprint time was improved significantly by $3.0 \pm 3.5\%$ ($P < 0.05$) in the light group. No significant differences were found between the groups in the changes in 5-m and 10-m sprint time from pre- to post-training. These results question the notion that training loads that induce greater than 10% reduction in sprint velocity would negatively affect sprint performance, and point out the potential benefit of using a heavier load for weighted sled towing.

Key Words: Ground Reaction Force; Resisted Sprint Training; Speed; Sprinting; Team Sport

INTRODUCTION

Sprint running speed is a fundamental fitness quality in many team sports such as soccer, rugby and basketball, and improving this fitness quality is one of the main goals of athletes' physical preparation in those sports (20). In particular, the ability to rapidly increase sprint velocity over short distances, which is often termed sprint acceleration ability, is considered to be of major importance, because sprint efforts are typically of short duration (e.g., 10-20 m, 2-3 s) during team-sport competition (20, 21).

Whereas actual practice of sprint skill is the most specific and essential method for enhancing sprint performance, other types of training such as weight training, plyometric training, flexibility training, and assisted sprint training are also used supplementary to development of sprint acceleration ability (7, 9, 23). Resisted sprint training is one such supplemental training method, in which athletes sprint with added loads such as uphill running, weighted sled towing, and weighted vest sprinting (5, 12). Weighted sled towing is a commonly used form of resisted sprint training, which uses mass (inertia) of a sled as well as the friction between the sled and the ground surface as external resistance to sprint movement (5, 22). It is assumed that directly adding an external load to sprint movement in the form of a weighted sled increases the demand for horizontal ground reaction force (GRF) impulse production at each ground contact. Repetitively applying such a stimulus over time is likely to induce specific adaptations within the neuromuscular system allowing larger horizontal GRF impulse production, which would theoretically lead to increased step length, thereby improving sprint performance under un-resisted conditions (5, 15, 22).

Despite the popularity of weighted sled towing and seemingly logical assumption behind its mechanisms, little experimental evidence exists to support its effectiveness. To the best of our knowledge, only three peer-reviewed studies have reported the effects of training with weighted sled towing on sprint acceleration ability. Zafeiridis et al. (24) compared the effects of 8 wk of weighted sled towing with 5 kg and un-resisted sprint training, and found that the sled towing group significantly improved sprint acceleration performance (0-20 m), whereas the un-resisted group did not. Spinks et al. (22) conducted an 8-wk training study comparing the effects of weighted sled towing with 12.6% of body mass versus un-resisted sprint training. They reported that the weighted sled towing significantly improved sprint acceleration performance (0-15 m) compared with a non-training control group, but the improvement was not significantly different from that of un-resisted sprint training. Harrison and Bourke (11) examined the effect of adding weighted sled towing with 13% of body mass to normal rugby training, which included (un-resisted) speed training drills, over 6 wk. They found that the addition of weighted sled towing to the rugby training improved sprint acceleration performance (0-5 m) to a greater extent than normal rugby training alone. Viewed collectively, the limited existing literature indicates that short-term training (≤ 8 wk) with weighted sled towing is more effective than or at least as effective as un-resisted sprint training for improving sprint acceleration performance.

Controversy exists as to the optimal training load of weighted sled towing for enhancing sprint ability (5). Coaches and sports scientists generally recommend that athletes

perform weighted sled towing with relatively light external resistance, which is $\leq 10\%$ of body mass, or which reduces sprint velocity by $\leq 10\%$ (1, 12, 14, 15, 22). The rationale behind this so called “10% rule” is the belief that applying heavy external resistance ($> 10\%$) to sprint movement would alter sprint mechanics excessively, and the long-term use of such heavy resistance results in chronic deterioration of sprint technique and performance under un-resisted conditions (1, 15). It is also argued that the use of light training loads is necessary to ensure the specificity of weighted sled towing (1, 12, 15). However, this line of reasoning against the use of a heavy training load for weighted sled towing is largely based on coaches’ subjective opinions, and no research has provided evidence to support or refute it (12, 14, 15). Moreover, it could be argued that performing weighted sled towing with a light load may not provide sufficient contrast to practicing sprint skill under un-resisted conditions, questioning the use of external resistance at all.

No previous research has compared the training effects of weighted sled towing with different magnitudes of external resistance in a single study and specifically tested the “10% rule”. Therefore, the purpose of this study was to compare the training effects of weighted sled towing with two different external loads that reduced sprint velocity by $\sim 10\%$ and $\sim 30\%$, and to determine the effect of the magnitude of a training load in weighted sled towing on the development of sprint acceleration ability. We hypothesized that the use of heavier load would provide greater training overload and improve sprint acceleration performance to a greater extent than the light load. Information derived

from this study should help validate or dismiss the “10% rule” for weighted sled towing, and would allow coaches to better prescribe weighted sled towing for their athletes.

METHODS

Experimental Approach to the Problem

This study used a two-group, randomized, longitudinal experimental design to compare the training effects of weighted sled towing between heavy and light loads. All subjects attended a baseline testing session consisting of a 10-m sprint test before the initiation of the training intervention. Subsequently, the subjects were matched for 10-m sprint time in the pre-test and allocated to either a heavy-load weighted sled towing group (heavy group; $n = 10$) or a light-load weighted sled towing group (light group; $n = 11$). Each group had similar sprint acceleration ability at baseline (Table 1). As the independent variable was the “magnitude of external load”, we did not include any control group (either un-resisted sprint training group or no sprint training group) in this study. Within 7 days of the completion of their pre-training test, all subjects commenced a supervised 8-wk training program, in which the subjects in the heavy and light groups performed weighted sled towing twice per week using external loads that reduced 10-m sprint velocity by ~30% and ~10%, respectively. Within 10 days of the completion of the 8-wk training intervention, the subjects repeated the same test as performed prior to the training, and then changes in the dependent variables before and after training were compared between the groups.

Subjects

Twenty-one physically active men who had experience in team sports such as soccer, basketball, field hockey, and Australian Rules football were recruited for this study. Their performance standards ranged from recreational performers to regional-level athletes. When this study was conducted, the subjects were either in their off-season or pre-season preparation period, and none of them were in the competition period. The sample size was determined based on the previous studies that examined the effects of weighted sled towing on sprint acceleration performance (11, 22, 24). The subjects' characteristics are shown in Table 1. The subjects were instructed to maintain their normal diet and training pattern for the duration of the study.

(Table 1 about here)

This project was reviewed and approved by the institutional ethics committee. All the subjects were informed of the study requirements, benefits, and possible risks, and gave their written informed consent before the study began. From two of the subjects who were under 18 y, informed consent was obtained also from their parents/guardians.

Training Procedures

Training Intervention

The subjects participated in two supervised sprint training sessions per week (on non-consecutive days) for 8 wk, which were performed on an outside grassed field. Each training session began with a thorough warm-up consisting of light jogging (5 min), dynamic stretching exercises (~10 min), and submaximal un-resisted 10-m sprints at 50, 75, and 90% of maximal intensity, each performed for one repetition. Subjects then

performed a series of maximal-effort weighted sled towing over 5-15 m with 1-2 min of rest between repetitions (Table 2). The sprint distances of 5 and 15 m were included in the training program in addition to 10 m, to add variety to the program and so that the subjects could work technically on the early acceleration (5 m) and the longer acceleration (15 m). The sprint training program followed a periodized plan to allow progression and to prevent overtraining. The total number of sprints and the total sprint distance covered were matched between the groups. However, the external loads added were different between the groups such that the heavy group used a weighted sled that reduced 10-m sprint velocity by ~30% relative to the un-resisted condition, whereas the light group used a weighted sled that reduced 10-m sprint velocity by ~10%. The selection of this training load for the light group was based on what coaches and sports scientists generally recommend as the optimal training load of weighted sled towing (14, 15). On the other hand, the training load for the heavy group was chosen arbitrarily to examine the training effect of weighted sled towing with an external load that is greater than what is typically recommended. Note that we use the terms “light” and “heavy” in this study to describe the magnitudes of the two added loads in a relative (rather than absolute) sense. During weighted sled towing, the participants were connected to the sled via a shoulder harness that was in turn connected to the sled by a connecting lead.

(Table 2 about here)

Training Monitoring and Sled Mass Adjustment

In the first training session of the 8-wk training period, the absolute sled mass for each subject was selected based on his body mass, referring to the method of Lockie et al. (15).

In the subsequent training sessions, the sled mass was adjusted individually using the following procedure: (a) At the end of a warm up, the subjects performed three maximal un-resisted 10-m sprints without towing a sled, which were timed using timing lights (Swift Performance Equipment, Lismore, Australia); (b) The subjects completed the prescribed weighted sled towing program as shown in Table 2, during which 10-m sprint time was recorded under resisted conditions using timing lights; (c) After the training session, the 10-m sprint velocity under un-resisted and resisted conditions were compared, and the absolute sled mass was adjusted individually for the subsequent training session so that the subjects always sprinted at the prescribed percentages of maximal un-resisted sprint velocity (i.e., ~70% and ~90% for the heavy and light groups, respectively). This adjustment procedure was performed once a week for a total of eight times over the 8-wk training intervention period.

Testing Procedures

Before and after the 8-wk training period, the subjects performed a 10-m sprint test. The time of day for testing was standardized (± 2 h) on each testing occasion, and the subjects were instructed to refrain from strenuous exercise for 24 h before each testing session. The subjects were also requested to be rested, normally hydrated, and abstain from eating 2 hours prior to testing. During testing and training strong verbal encouragement was provided to increase motivation levels.

10-m Sprint Test

This test was used to quantify sprint acceleration ability. Although “acceleration” is physically defined as the rate of change in velocity, we used 10-m sprint time as a working/practical definition of sprint acceleration ability in this study (18). The test distance of 10 m was chosen because: (a) it is often used to assess sprint acceleration ability of team-sport athletes (2, 10); (b) mean sprint distances during team-sport competitions are typically 10-20 m (21); and (c) the initial 10 m of sprint running has been shown to be a specific component representing initial acceleration ability (8).

The 10-m sprint test was conducted indoors on a hard flat surface. Prior to the sprint trials, the subjects undertook a standardized warm-up that involved light jogging, dynamic stretching exercises, and familiarization with the sprint protocol at submaximal intensity. Subjects then performed six maximal-effort sprints over 10 m from a parallel standing start (6); the subjects started in a standing position with the toes of both feet parallel at 0.3 m behind the initial timing gate and with the first step moving forward (no step backward was allowed). Rest periods of ~2 min were allowed between trials to ensure a maximal effort and minimum fatigue for each sprint (3). The split time at 5 and 10 m from the start line was measured using a dual-beam electronic timing system (Swift Performance Equipment, Lismore, Australia), with an accuracy of 0.01 s. The mean sprint time of the three best trials was used for statistical analyses.

Ground Reaction Force Analyses

During the 10-m sprint tests, ground reaction forces (GRF) were collected at a sampling frequency of 1000 Hz, separately at the first ground contact and at 8 m from the start

using three force plates (Type 9287BA, Kistler Instrument Corp., Winterthur, Switzerland) recessed in the floor, giving a total length of 2.7 m measurement surface. Since it was not possible for us to simultaneously record GRF at the two different points in the same trial, GRF at the first ground contact after the start was collected in the first three trials, then we changed the start line and the placement of the timing gates, and GRF at 8 m from the start was recorded in the subsequent three trials. The recorded GRF data were filtered using a fourth-order recursive, zero phase-shift Butterworth low-pass filter with a cut-off frequency of 100 Hz. Then, we calculated resultant, vertical (effective), net horizontal, propulsive, and braking impulses, using methods described by Kawamori et al. (13), and all the GRF impulses were normalized to body mass (17). The GRF variables from the three trials were averaged separately for each recording point to be used for statistical analyses.

Video Analyses

During the 10-m sprint tests, sagittal plane video data were recorded at the first ground contact and at 8 m from the start using a digital video camera (MV920, Canon Inc., Ohtaku, Japan) that was placed perpendicular to the sagittal plane of motion at a distance of 6 m from the midline of the running lane. The shutter speed of the video camera was 1/500 s, and the camera view was calibrated using a frame of known dimension. As already described, video at the first ground contact after the start was recorded in the first three trials, and then video at 8 m from the start was recorded in the subsequent three trials. To analyze the recorded sagittal plane video data, video analysis software (siliconCOACH Pro, siliconCOACH Ltd., Dunedin, New Zealand) was used. Step length was the

horizontal distance between the point of touchdown of one foot and that of the following touchdown of the opposite foot, and step frequency was the reciprocal of step time (i.e., time taken to complete a step). Only one step was analyzed for each point (i.e., the first ground contact and 8 m). It should be noted that although step cycles analyzed are those “around” the first ground contact and 8 m rather than exactly “at” the first ground contact and 8 m, the expression “at” will be used for simplicity. Similarly, the expression “GRF recorded at 8 m” will be used although it is “GRF recorded around 8 m” to be exact. Video-derived variables from the three trials were averaged separately for each recording point and used for statistical analyses.

The reliability of measures derived from the 10-m sprint test (time, GRF) was determined in our previous study (13) using a similar group of subjects, and the same investigator. The intraclass correlation coefficient (ICC) for 10-m sprint time and GRF ranged between 0.56 and 0.74, and 0.57 and 0.83, respectively.

Statistical Analyses

An independent t test was performed on the baseline measures to determine if the two groups were evenly matched before the training intervention. A two-way repeated measures ANOVA with “group” (heavy, light) as the between-subject factor and “time” (pretest, posttest) as the within-subject factor was used on each dependent variable, to examine difference in the change over time between heavy and light groups. When a significant F value was found, LSD post hoc tests were performed. The level of statistical significance was set at $P < 0.05$. Statistical analyses were performed using the

software package SPSS version 16.0 (SPSS Inc., Chicago, USA). All raw data are shown as mean $\pm$ SD.

RESULTS

Pre-training Testing

No significant differences between the groups were evident for any variables before the training (Tables 3 - 5).

(Tables 3, 4, 5 about here)

Training Analyses

Training Compliance

There was no difference in the number of training sessions attended by each subject between the groups over the course of the study (heavy = 84%, light = 84%). Missed training sessions were rescheduled, but complete compliance could not be achieved because of the reasons such as minor injuries (not related to the experiment), illness, weather (rain), travel, and competition commitments.

Training Monitoring and Sled Mass Adjustment

Over the 8-wk training period, the subjects in the heavy and light groups used sleds weighing 33.1 ± 5.9 kg and 10.8 ± 2.3 kg, corresponding to $43.0 \pm 7.0\%$ and $13.1 \pm 2.1\%$ of their body mass, respectively. The use of such sled mass resulted in decrements in 10-m sprint velocity by $28.1 \pm 1.5\%$ (heavy) and $9.7 \pm 1.3\%$ (light). How some of these variables changed over the course of the 8-wk training period is presented in Figure 1.

(Figure 1 about here)

Effects of Weighted Sled Towing

10-m Sprint Test

10-m sprint time significantly decreased from pre- to post-training by $5.0 \pm 3.5\%$ in the heavy group ($p < 0.001$) and by $3.0 \pm 3.5\%$ in the light group ($p = 0.008$) (Table 3). No significant group $\times$ time interaction was found for 10-m sprint time. Over the first 5 m, sprint time significantly decreased from pre- to post-training by $5.7 \pm 5.7\%$ in the heavy group ($p = 0.004$) but the $2.8 \pm 4.9\%$ change in the light group did not reach statistical significance. No significant group $\times$ time interaction was found in 5-m sprint time.

Ground Reaction Force Analyses

At the first ground contact after the start, no significant group $\times$ time interaction was observed for relative impulses (Table 4). Similarly, there were no significant changes from pre- to post-training in relative impulses.

At 8 m from the start, significant group $\times$ time interactions were found for resultant impulse and vertical impulse, but not for the other impulse measures (Table 5). The heavy group significantly decreased resultant impulse by $4.3 \pm 6.7\%$ ($p = 0.019$) and vertical impulse by $11.5 \pm 13.7\%$ ($p = 0.005$) from pre- to post-training, whereas the light group did not. The other GRF-related measures did not change significantly from pre- to post-training in either group, with no significant group $\times$ time interactions.

Video Analyses

At the first ground contact after the start, no significant group $\times$ time interaction was found for step frequency and step length (Table 4). The heavy group significantly increased step length from pre- to post-training by $8.1 \pm 10.8\%$, whereas the light group showed no significant change. Step frequency did not change significantly from pre- to post-training in either group.

At 8 m from the start, no significant group $\times$ time interaction was observed for step frequency and step length (Table 5). The heavy group significantly increased step frequency from pre- to post-training by $8.1 \pm 9.0\%$, whereas the light group showed no significant change. Step length increased significantly from pre- to post-training by $5.1 \pm 7.3\%$ in the light group, but not in the heavy group.

DISCUSSION

The main finding of this study is that 8-wk training programs of weighted sled towing with heavy or light training loads that reduced sprint velocity by $\sim 30\%$ and $\sim 10\%$ were similarly effective in improving 10-m sprint performance. It is often recommended that weighted sled towing be performed using relatively light external resistance such as a load that reduces sprint velocity by $\leq 10\%$ (i.e., 10% rule), based on the assumption that adding too heavy external resistance to sprint movement would disrupt running mechanics and eventually lead to decreased sprint performance (15). However, the lack of significant difference observed in this study between heavy and light groups in the pre-post changes in 10-m sprint performance does not support such a contention. In fact, 5-m

sprint performance improved only in the heavy group, indicating the potential advantage of using a heavier sled in developing sprint performance in the initial acceleration phase when loads are high and movement speed is relatively lower.

Theoretically, the heavier the added load in weighted sled towing, the greater the training overload, but the less specific the training movement becomes to un-resisted sprint running. It may be that the balance between “overload” and “specificity” in a training stimulus will ultimately determine the training outcome, and the training load that optimizes such a balance would maximize the training effect of weighted sled towing. Alternatively, it could be argued that weighted sled towing is effective in improving sprint performance not because it is specific to sprint movement but rather the neuromuscular adaptations it causes are what is required for improving sprint acceleration performance. If this is the case, then the fact that weighted sled towing is movement specific to normal sprint movement is irrelevant, and the fact that adding heavier load in weighted sled towing brings about larger acute changes in sprint kinematics (15, 19) should not be a concern in selecting training load for weighted sled towing. Either way, we examined only two training loads in this study, and thus it is unknown whether the use of even lighter or heavier external resistance (e.g., loads that reduce sprint velocity by ~5% or ~50%) improves sprint acceleration ability to a greater extent. Thus, additional research examining various loads other than those used in this study is required to identify the true optimal training load of weighted sled towing.

Regarding the video and GRF analyses of the 10-m sprint, the most notable finding is that the heavy group significantly decreased resultant and vertical impulses at 8 m from pre- to post-training and such changes were significantly larger than those in the light group. On the other hand, horizontal components of impulses did not show any significant changes in either group. These findings are unexpected because we originally hypothesized that training with weighted sled towing would improve impulse production especially in a horizontal direction, which would in turn increase step length, and that such changes would be magnified by using heavier resistance. This hypothesis, especially the former part, is common among coaches and sports scientists (15), but not supported by our findings. On the contrary, the improvement in 10-m sprint time by heavy-load weighted sled towing in this study was accompanied by the increased step frequency, which may be attributed to decreased vertical impulse production, both observed at 8 m from the start. Based on such findings, it seems necessary to revise our understanding of potential mechanisms of weighted sled towing to improve sprint performance; that is, weighted sled towing especially with heavy loads improves sprint acceleration performance by teaching athletes to direct GRF impulse more horizontally, not necessarily by allowing athletes to produce larger horizontal or resultant GRF impulse. In fact, recent biomechanical research shows that the orientation or the angle of GRF applied to the ground during the contact phase is more important to sprint acceleration performance than the total amount of GRF (13, 16), indicating that this is a favorable change for sprint acceleration. If this new hypothesis regarding the potential mechanism of weighted sled towing is correct, then weighted sled towing could be regarded as a skill practice exercise more than a strengthening exercise. That is, it does

not necessarily strengthen the neuromuscular systems involved in sprinting in a specific manner, as many believe, but rather it teaches a more efficient/effective way to apply GRF impulse while sprinting, which has more to do with motor control than muscular strength improvement per se. Alternatively, it is still possible that weighted sled towing does improve neuromuscular capacities involved in sprinting in a specific manner and it is these stronger and more powerful muscles that allow athletes to direct GRF impulse more horizontally, which was not possible with weaker muscles.

It is important to recognize the fact that the subjects in this study performed not only weighted sled towing but also a considerable volume of un-resisted sprints within our training program as a part of warm-up and sled mass adjustment procedure, and also outside our program during skill practice sessions and competitions. This perhaps allowed them to practice with their more efficient/effective motor pattern and/or stronger and more powerful muscles (developed by weighted sled towing) under un-resisted conditions, thereby potentially facilitating the transfer of training effects. If the subjects performed weighted sled towing only and nothing else, the results might have been different. However, such a training program (i.e., weighted sled towing only) is not realistic for competitive athletes, and this should not be a concern for coaches.

A unique feature of this study was that we prescribed the training load (i.e., sled mass) individually for each subject based on percent decrements in sprint velocity. Such a method is different from those of previous studies in which a training load was chosen as absolute sled mass (e.g., 5 kg) for all the subjects (24) or as percentages of body mass

(e.g., 12.6-15% of body mass) (11, 22). We believe our method was more appropriate because it allows the training load to be tailored for each athlete regardless of his/her body mass, sprint ability, muscular strength, or any other factors that may influence the degree of decrement in sprint velocity when towing a weighted sled. In addition, the training load selected this way is not affected by factors such as running surface and sled characteristics, which influence the magnitude of friction (force) and change the external resistance applied to an athlete, even when towing a sled of the same absolute mass. Such a sled mass adjustment and monitoring strategy may be perceived as too time-consuming and not practical, but worked well in this study. Moreover, a frequent measurement and feedback of sprint time has an added benefit of motivating athletes, which is worth the time and effort in actual training settings.

Before concluding, we must highlight several limitations/delimitations of this study. First, the results of this study are likely to be specific to the use of weighted sled towing for improving sprint performance in the acceleration phase. Thus, they may not apply to other types of resisted sprint training (e.g., uphill running, weighted vest sprinting) or other phases of sprint running (e.g., maximal speed, speed endurance). In fact, there exist a few research studies indicating that weighted sled towing may not improve maximal speed ability or longer sprint performance as much as un-resisted sprint training does (4, 24). Second, we did not control the subjects' involvement in other types of training/exercises outside the scope of this study (e.g., sports practice, weight training, endurance training) during the intervention period, although we asked the subjects to maintain their normal training pattern. It is possible that concurrently performing other

types of training affects the effect of weighted sled towing either positively or negatively, and therefore this issue needs to be addressed in future studies with tighter control. Third, although the total sprint training distance was matched between the groups, training volume in terms of mechanical work performed was most likely different (i.e., heavy > light) because towing a heavier sled requires greater work output. Therefore, any differences in change scores observed between the groups cannot be attributed solely to a load effect, and may be also due to the difference in the total work performed. Finally, the subjects in this study were either competitive (regional level) or recreational athletes. Thus, the applicability of the findings of this study to athletes of higher competition levels (e.g., elite or high-performance) may be limited.

In conclusion, both heavy- and light-load weighted sled towing were equally effective for improving 10-m sprint performance, but only heavy-load weighted sled towing improved 5-m sprint performance. Thus, our findings did not validate the typical coaching guideline of using relatively light load (i.e., 10% rule) and indicate the potential benefit of using a heavier load for weighted sled towing.

PRACTICAL APPLICATIONS

Heavy- and light-load weighted sled towing were equally effective in improving sprint acceleration ability over 10 m, but only the use of heavy load improved 5-m sprint performance. Therefore, it is conclusive that coaches and athletes should abandon the myth regarding the optimal training load of weighted sled towing (i.e., 10% rule) and should explore the use of heavier external resistance for weighted sled towing.

ACCEPTED

REFERENCES

1. Alcaraz PE, Palao JM, and Elvira JL. Determining the optimal load for resisted sprint training with sled towing. *J Strength Cond Res* 23: 480-485, 2009.

2. Baker D and Nance S. The relation between running speed and measures of strength and power in professional rugby league players. *J Strength Cond Res* 13: 230-235, 1999.
3. Balsom PD, Seger JY, Sjodin B, and Ekblom B. Maximal-intensity intermittent exercise: effect of recovery duration. *Int J Sports Med* 13: 528-533, 1992.
4. Clark KP, Stearne DJ, Walts CT, and Miller AD. The longitudinal effects of resisted sprint training using weighted sleds vs. weighted vests. *J Strength Cond Res* 24: 3287-3295, 2010.
5. Cronin J and Hansen KT. Resisted sprint training for the acceleration phase of sprinting. *Strength Cond J* 28: 42-51, 2006.
6. Cronin JB, Green JP, Levin GT, Brughelli ME, and Frost DM. Effect of starting stance on initial sprint performance. *J Strength Cond Res* 21: 990-992, 2007.
7. Delecluse C. Influence of strength training on sprint running performance. Current findings and implications for training. *Sports Med* 24: 147-156, 1997.
8. Delecluse C, Van Coppenolle H, Willems E, Van Leemputte M, Diels R, and Goris M. Influence of high-resistance and high-velocity training on sprint performance. *Med Sci Sports Exerc* 27: 1203-1209, 1995.
9. Dintiman GB. Effects of various training programs on running speed. *Res Q* 35: 456-463, 1964.
10. Duthie GM, Pyne DB, Ross AA, Livingstone SG, and Hooper SL. The reliability of ten-meter sprint time using different starting techniques. *J Strength Cond Res* 20: 246-251, 2006.

11. Harrison AJ and Bourke G. The effect of resisted sprint training on speed and strength performance in male rugby players. *J Strength Cond Res* 23: 275-283, 2009.
12. Jakalski K. The pros and cons of using resisted and assisted training methods with high school sprinters: parachutes, tubing, and towing. *Track Coach* 144: 4585-4589, 4612, 1998.
13. Kawamori N, Nosaka K, and Newton RU. Relationships between ground reaction impulse and sprint acceleration performance in team-sport athletes. *J Strength Cond Res* [Epub ahead of print].
14. Letzelter M, Sauerwein G, and Burger R. Resistance runs in speed development. *Mod Athlete Coach* 33: 7-12, 1995.
15. Lockie RG, Murphy AJ, and Spinks CD. Effects of resisted sled towing on sprint kinematics in field-sport athletes. *J Strength Cond Res* 17: 760-767, 2003.
16. Morin JB, Edouard P, and Samozino P. Technical Ability of Force Application as a Determinant Factor of Sprint Performance. *Med Sci Sports Exerc* 43: 1680-1688, 2011.
17. Mullineaux DR, Milner CE, Davis IS, and Hamill J. Normalization of ground reaction forces. *J Appl Biomech* 22: 230-233, 2006.
18. Murphy AJ, Lockie RG, and Coutts AJ. Kinematic determinants of early acceleration in field sport athletes. *J Sports Sci Med* 2: 144-150, 2003.
19. Murray A, Aitchison T, Ross G, Sutherland K, Watt I, McLean D, and Grant S. The effect of towing a range of relative resistances on sprint performance. *J Sports Sci* 23: 927-935, 2005.

20. Sayers M. Running techniques for field sport players. *Sports Coach* 23: 26-27, 2000.
21. Spencer M, Bishop D, Dawson B, and Goodman C. Physiological and metabolic responses of repeated-sprint activities: specific to field-based team sports. *Sports Med* 35: 1025-1044, 2005.
22. Spinks CD, Murphy AJ, Spinks WL, and Lockie RG. The effects of resisted sprint training on acceleration performance and kinematics in soccer, rugby union, and Australian football players. *J Strength Cond Res* 21: 77-85, 2007.
23. Young W, Benton D, Duthie G, and Pryor J. Resistance training for short sprints and maximum-speed sprints. *Strength Cond J* 23: 7-13, 2001.
24. Zafeiridis A, Saraslanidis P, Manou V, Ioakimidis P, Dipla K, and Kellis S. The effects of resisted sled-pulling sprint training on acceleration and maximum speed performance. *J Sports Med Phys Fitness* 45: 284-290, 2005.

ACKNOWLEDGEMENTS

We thank David Bell for his help with data collection. We also thank Jonathon Green for his technical assistance. There was no financial assistance with the project.

FIGURE LEGENDS

FIGURE 1. Training monitoring and sled mass adjustment: (a) changes in sled mass, expressed as percent of body mass, over the 8-wk training period; and (b) corresponding changes in 10-m sprint velocity during weighted sled towing, expressed as percent of unresisted 10-m sprint velocity, over the 8-wk training period.

ACCEPTED

Table 1. Characteristics of the subjects in the two training groups.*

	Heavy group (<i>n</i> = 10)	Light group (<i>n</i> = 11)
Sample size by level of sports participation		
Competitive	5	8
Recreational	5	3
Characteristics		
Age (yr)	22.8 ± 3.3	22.3 ± 5.2
Height (m)	1.79 ± 0.08	1.83 ± 0.07
Body mass (kg)	77.5 ± 7.3	82.7 ± 9.0
10-m sprint time (s)	2.05 ± 0.07	2.06 ± 0.10
Concurrent involvement in weight training (% of sample)	80%	73%

*Data were taken before the 8-wk training period, and are expressed as mean ± SD.

Table 2. Sprint training program.

Week	Repetition × distance			Total number of	Total distance
				repetitions	(m)
1	3 × 5 m	3 × 10 m	3 × 15 m	9	90
2	3 × 5 m	4 × 10 m	3 × 15 m	10	100
3	3 × 5 m	5 × 10 m	3 × 15 m	11	110
4	3 × 5 m	3 × 10 m	3 × 15 m	9	90
5	4 × 5 m	4 × 10 m	4 × 15 m	12	120
6	4 × 5 m	5 × 10 m	4 × 15 m	13	130
7	4 × 5 m	6 × 10 m	4 × 15 m	14	140
8	3 × 5 m	3 × 10 m	3 × 15 m	9	90

Table 3. Pre- and post-test results for 5- and 10-m sprint time during the 10-m sprint test for heavy and light groups.

Variable	Heavy group (<i>n</i> = 10)		Light group (<i>n</i> = 11)		Interaction
	Pretest	Posttest	Pretest	Posttest	P value
5-m time (s)	1.28 ± 0.06	1.21 ± 0.07*	1.28 ± 0.08	1.24 ± 0.07	0.260
10-m time (s)	2.05 ± 0.07	1.94 ± 0.08*	2.06 ± 0.10	2.00 ± 0.10*	0.151

*Significant difference from pretest at $P < 0.05$

Table 4. Pre- and post-test results for biomechanical variables recorded at the first ground contact after the start during the 10-m sprint test for heavy and light groups.

Variable	Heavy group (<i>n</i> = 10)		Light group (<i>n</i> = 11)		Interaction
	Pretest	Posttest	Pretest	Posttest	P value
Sagittal-plane video data					
Step frequency (Hz)	3.46 ± 0.34	3.46 ± 0.39	3.48 ± 0.35	3.50 ± 0.40	0.870
Step length (m)	1.03 ± 0.12	1.11 ± 0.12*	1.05 ± 0.10	1.10 ± 0.11	0.537
Ground reaction force data					
Relative resultant impulse (m·s ⁻¹)	2.98 ± 0.30	2.99 ± 0.37	2.90 ± 0.23	2.88 ± 0.33	0.780
Relative vertical impulse (m·s ⁻¹)	0.22 ± 0.19	0.20 ± 0.18	0.22 ± 0.15	0.17 ± 0.19	0.436
Relative net horizontal impulse (m·s ⁻¹)	1.20 ± 0.32	1.16 ± 0.52	1.03 ± 0.29	1.00 ± 0.47	0.926
Relative braking impulse (m·s ⁻¹)	-0.01 ± 0.01	-0.02 ± 0.03	-0.01 ± 0.01	-0.01 ± 0.01	0.379
Relative propulsive impulse (m·s ⁻¹)	1.21 ± 0.32	1.18 ± 0.51	1.04 ± 0.29	1.01 ± 0.47	0.966

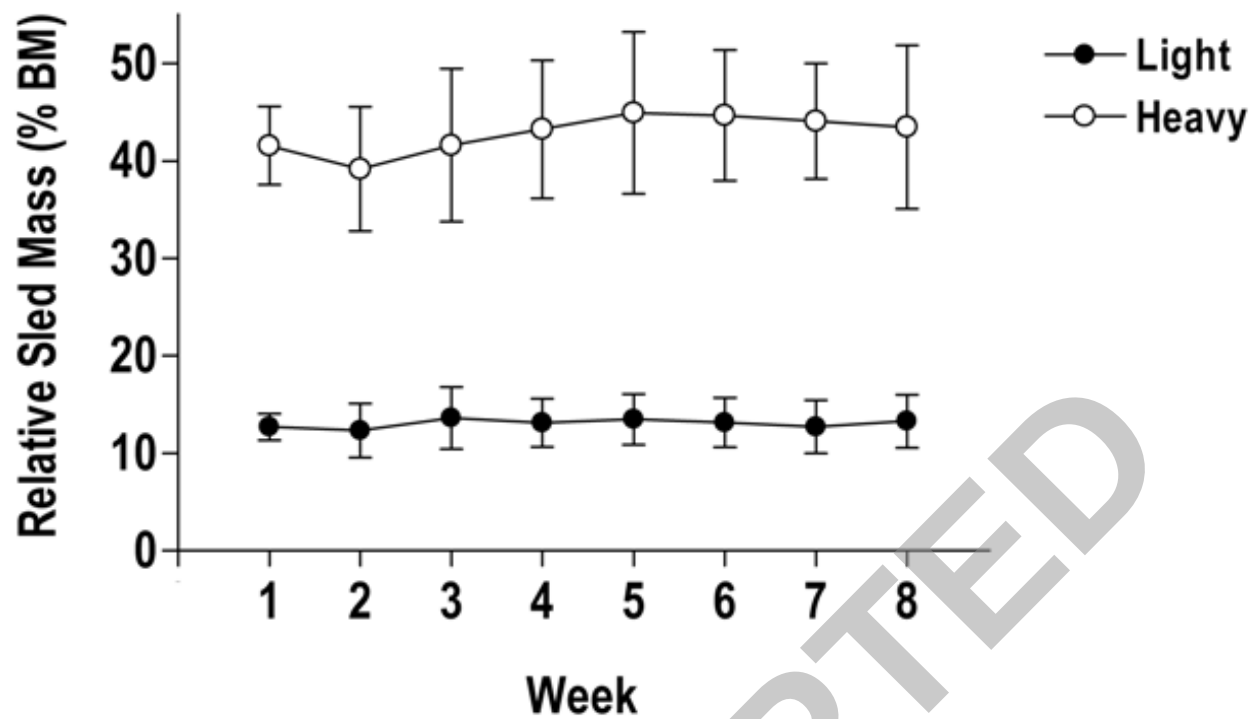
*Significant difference from pretest at *P* < 0.05

Table 5. Pre- and post-test results for biomechanical variables recorded at 8 m from the start during the 10-m sprint test for heavy and light groups.

Variable	Heavy group (<i>n</i> = 10)		Light group (<i>n</i> = 11)		Interaction
	Pretest	Posttest	Pretest	Posttest	P value
Sagittal-plane video data					
Step frequency (Hz)	4.02 ± 0.40	4.33 ± 0.36*	4.06 ± 0.31	4.11 ± 0.37	0.160
Step length (m)	1.68 ± 0.13	1.72 ± 0.15	1.64 ± 0.09	1.72 ± 0.14*	0.390
Ground reaction force data					
Relative resultant impulse (m·s ⁻¹)	2.51 ± 0.22	2.39 ± 0.11*	2.45 ± 0.15	2.49 ± 0.18	0.023†
Relative vertical impulse (m·s ⁻¹)	0.92 ± 0.14	0.80 ± 0.06*	0.81 ± 0.11	0.82 ± 0.14	0.020†
Relative net horizontal impulse (m·s ⁻¹)	0.39 ± 0.06	0.41 ± 0.05	0.37 ± 0.06	0.37 ± 0.08	0.439
Relative braking impulse (m·s ⁻¹)	-0.06 ± 0.02	-0.06 ± 0.02	-0.07 ± 0.02	-0.08 ± 0.02	0.254
Relative propulsive impulse (m·s ⁻¹)	0.45 ± 0.05	0.47 ± 0.04	0.44 ± 0.06	0.45 ± 0.07	0.812

*Significant difference from pretest at $P < 0.05$; †significant group × time interaction at $P < 0.05$

(a)



(b)

